

Acids and Bases

Friday, 5 June 2020 9:48 AM

Acids

- Bronsted Lowry: A proton donor
- Arrhenius: A substance which ionises to form H^+ ions in aqueous solution
- Turns blue litmus red, sour taste
- Many non-metal oxides are acidic (combines with $\text{H}_2\text{O}_{(l)}$ to form acid)
 - E.g. $\text{CO}_{2(g)} + \text{H}_2\text{O}_{(l)} \rightleftharpoons \text{H}_2\text{CO}_{3(aq)}$
 - $\text{H}_2\text{CO}_{3(aq)} \rightleftharpoons 2\text{H}^+_{(aq)} + \text{CO}_3^{2-}$

Strong Acids

- Ionise completely, single arrow
- 8 strong acids:

Hydrochloric Acid	HCl
Hydrobromic Acid	HBr
Hydroiodic Acid	HI
Nitric Acid	HNO_3
Sulfuric Acid	H_2SO_4
Chloric Acid	HClO_3
Perchloric Acid	HClO_4
Periodic Acid	HIO_4

Weak Acids

- Ionise partially, double arrow
- If e.g. 1000 000 molecules dissolved in water, only a few will ionise
- When rate of forward reaction is equal to rate of reverse reaction, equilibrium is achieved
- If not a strong acid, it is a weak acid
- In ionic equations, leave as a molecule as it is mostly present as molecules

Monoprotic vs Polyprotic

- Monoprotic acids donate one proton per molecule
- Polyprotic acids donate multiple protons per molecule
 - Ionise in multiple stages
 - E.g. $\text{H}_2\text{SO}_{4(aq)} \rightarrow \text{H}^+_{(aq)} + \text{HSO}_4^-_{(aq)}$
 - $\text{HSO}_4^-_{(aq)} \rightarrow \text{H}^+_{(aq)} + \text{SO}_4^{2-}_{(aq)}$
 - Overall, $\text{H}_2\text{SO}_{4(aq)} \rightarrow \text{H}^+_{(aq)} + \text{SO}_4^{2-}_{(aq)}$

Acidity

- Acidity: The ability of an acid to donate protons
- In order of most to least acidic: H_2SO_4 , HCl, H_3PO_4
 - H_2SO_4 donates 2 protons per molecule
 - HCl donates 1 proton per molecule
 - H_3PO_4 is weak, we can't say how many protons it donates

Bases

- Bronsted Lowry: A proton acceptor
- Arrhenius: A substance which dissociates to form OH^- ions in aqueous solution
- Alkali: A base that is soluble in water
- Many metal oxides are basic (releases O^{2-} which combines with H_2O)
 - E.g. $\text{Na}_2\text{O}_{(s)} + \text{H}_2\text{O}_{(l)} \rightarrow 2\text{Na}^+_{(aq)} + 2\text{OH}^-_{(aq)}$

Strong Bases

- Group 1&2 hydroxides are strong bases
- Dissociate completely
- Soluble bases will be stronger than non-soluble bases as they can release more hydroxide ions into aqueous solution

Weak Bases

- All other bases are weak
- Ionise/Dissociate partially (double arrow)
- NH_3 is a weak base - although not hydroxide, it accepts protons
 - $\text{NH}_{3(\text{aq})} + \text{H}_2\text{O}_{(\text{l})} \rightleftharpoons \text{NH}_4^+_{(\text{aq})} + \text{OH}^-_{(\text{aq})}$

Reactions Involving Acids/Bases

Equations of Acidic Solutions

- Acid + Reactive Metal $\rightarrow$ Salt + Hydrogen Gas
- Acid + Metal Oxide/Hydroxide $\rightarrow$ Salt + Water
- Acid + Carbonate/Hydrogen carbonate $\rightarrow$ Salt + Water + Carbon Dioxide
 - This is a two-stage process: Initially forms carbonic acid (H_2CO_3) which decomposes into $\text{H}_2\text{O}_{(\text{l})} + \text{CO}_{2(\text{g})}$
- Acid + Metal Sulfite $\rightarrow$ Sulfur Dioxide ($\text{SO}_{2(\text{g})}$) + Salt + Water

Equations of Basic Solutions

- Base + Ammonium Salt $\rightarrow$ Ammonia Gas + Salt + Water
- Base + Non-Metal Oxide $\rightarrow$ Salt + Water

pH

- $\text{pH} = -\log_{10}[\text{H}^+]$; $[\text{H}^+] = 10^{-\text{pH}}$
- $[\text{H}^+] = \text{H}^+$ concentration in mol L^{-1}
- Acidic: $\text{pH} < 7$; Neutral: $\text{pH} = 7$; Basic: $\text{pH} > 7$
- $\text{pOH} = -\log_{10}[\text{OH}^-]$
- Use 2 decimals for pH (minimum)
- pH of strong acids usually 0-3
- pH of weak acids usually 4-6

Water

- Amphoteric: A substance that can act as both an acid and a base
- Water is amphoteric (acts as an acid when reacting with a base and a base when reacting with an acid)

Self Ionisation of Water

- Pure water contains $1 \times 10^{-7} \text{ mol L}^{-1} \text{H}^+$ and OH^- ions at 25°C
 - $2\text{H}_2\text{O}_{(\text{l})} \rightarrow \text{H}_3\text{O}^+_{(\text{aq})} + \text{OH}^-_{(\text{aq})}$
- K_w (Equilibrium constant) = $[\text{H}^+][\text{OH}^-]$
 - K_w of water at $25^\circ\text{C} = 1 \times 10^{-14}$
- $[\text{H}^+] = K_w / [\text{OH}^-]$
 - Therefore we can find pH if given the OH^- concentration

Class

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Acids

- Proton donors (Bronsted Lowry)
 - E.g. $\text{HCl}_{(\text{aq})}$
 - H^+ is just a proton; Call it a proton if writing, write as H^+ in equation
 - $\text{H}^+_{(\text{aq})} + \text{H}_2\text{O}_{(\text{l})} \rightarrow \text{H}_3\text{O}^+$ (Hydronium ion)
 - $\text{HCl}_{(\text{aq})} + \text{H}_2\text{O}_{(\text{l})} \rightarrow \text{H}_3\text{O}^+_{(\text{aq})} + \text{Cl}^-_{(\text{aq})}$

Strong vs Weak Acids

- Strength has NOTHING TO DO WITH CONCENTRATION

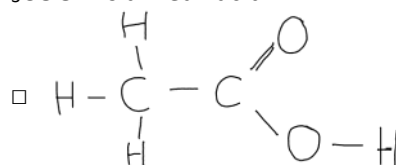
Strong Acids

- Strong acids **ionise completely** (almost every molecule will ionise)
- Single arrow** used to show reaction to completion
 - E.g. $\text{HCl}_{(\text{aq})} + \text{H}_2\text{O}_{(\text{l})} \rightarrow \text{H}_3\text{O}^+_{(\text{aq})} + \text{Cl}^-_{(\text{aq})}$
- 8 Strong Acids (the rest are weak):

Hydrochloric Acid	HCl
Hydrobromic Acid	HBr
Hydroiodic Acid	HI
Nitric Acid	HNO_3
Sulfuric Acid	H_2SO_4
Chloric Acid	HClO_3
Perchloric Acid	HClO_4
Periodic Acid	HIO_4

Weak Acids

- Weak acids **ionise partially**
- Reversible Reaction shown using **double arrow** ($\rightleftharpoons$)
 - Top arrow: Forward reaction
 - Bottom arrow: Reverse reaction
- When the **rate of the forward reaction** is **equal** to the **rate of the reverse reaction** **equilibrium** is achieved
- CH_3COOH is a weak acid



- The weakest bonded hydrogen atom will be removed
- $\text{CH}_3\text{COOH} + \text{H}_2\text{O}_{(\text{l})} \rightleftharpoons \text{CH}_3\text{COO}^-_{(\text{aq})} + \text{H}_3\text{O}^+$
- If e.g. 1000 000 molecules put in water, only a few of them will ionise
- Includes:
 - All acids containing carbon except H_2CO_3

Phosphoric Acid	H_3PO_4
Ethanoic Acid	CH_3COOH
Hydrofluoric Acid	HF
Hydrogen Sulfide	H_2S
Hydrocyanic Acid	HCN
Formic Acid	HCOOH

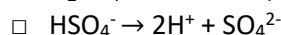
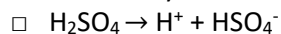
Monoprotic vs Polyprotic Acids

Monoprotic Acids

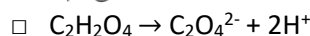
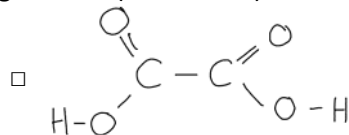
- Any acid which donates one proton per molecule is monoprotic
- E.g. HCl, HNO₃, CH₃COOH

Polyprotic Acids

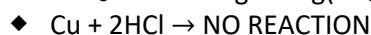
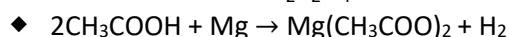
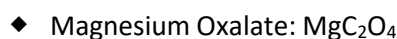
- Donates multiple protons per molecule
- Don't ionise fully in one step, multiple stages



- E.g. C₂H₂O₄ (Oxalic Acid)



- ☐ Practice:



- ◇ Anything above "0 EXACTLY" on half-reaction data sheet will not react with acids

Shen-Kit at 22/06/2020 1:44 PM

Acidity

- Most acidic to least acidic H₂SO₄, HCl, HCOOH
 - Acidity refers to the ability of an acid to donate protons
 - H₂SO₄ donates ~2 moles of protons per mole of acid
 - HCl donates ~1 mole of protons per mole of acid
 - HCOOH is a weak acid, we can't say how many protons it donates
- $\text{H}^+_{(\text{aq})} + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+$
- $\text{HCOOH}_{(\text{aq})} + \text{H}_2\text{O}_{(\text{l})} \rightarrow \text{H}_3\text{O}^+_{(\text{aq})} + \text{HCOO}^-_{(\text{aq})}$
- Water acts as a base when reacting with acids

Base

Definitions

- Bronsted Lowry: Substances that accept protons
- Arrhenius: Substance that creates hydroxide ions in aqueous solution (not all bases are soluble, doesn't work)
- Other: Bases taste bitter
- An **alkali** is a base which can dissolve in water

Strong vs Weak Bases

Strong

- Group 1&2 hydroxides** are strong bases
 - E.g. NaOH, Ca(OH)₂, Ba(OH)₂
 - Soluble bases will be stronger than non-soluble bases as they can release more hydroxide ions into aqueous solution and also therefore accept more protons
 - E.g. Ba(OH)₂ is stronger than Ca(OH)₂

Weak

- All other bases are weak bases
- NH₃ is a weak base - although it isn't a hydroxide, it accepts protons
 - $\text{NH}_3 + \text{H}_2\text{O} \rightleftharpoons \text{NH}_4^+_{(\text{aq})} + \text{OH}^-_{(\text{aq})}$
 - Water acts as an acid when reacting with bases
 - Amphoteric: A substance that can act as both an acid and a base
 - As water can act as an acid or a base (above), we say it is amphoteric

pH

- $pH = -\log[H^+]$
- $[H^+] = 10^{-pH}$
- Acidity of an aqueous solution depends on the number of H^+ (H_3O^+) ions in solution
- $pH < 7$: Acidic - high $[H^+]$
- $pH = 7$: Neutral
- $pH > 7$: Basic - high $[OH^-]$
- $[\]$ represents concentration
- Generally use 2 decimals for pH (no less)
- Strong acids have pH usually 0-3
- Weak acids have pH usually 4-6

Calculating $[H^+]$ from $[OH^-]$

- $H_2O(l) + H_2O(l) \rightleftharpoons H_3O^+_{(aq)} + OH^-_{(aq)}$
 - At 25°C: $10^{-7} \text{ mol L}^{-1} H_3O^+_{(aq)}$ and $10^{-7} \text{ mol L}^{-1} OH^-_{(aq)}$
- K_w (equilibrium constant of water) = $[H^+][OH^-] = 10^{-14}$ (in water at 25°C)
- If $[OH^-]$ is known, $[H^+] = K_w / [OH^-]$

HIA Shen-Kit at 1/07/2020 1:26 PM

- $0.1 \text{ mol L}^{-1} \text{ HCl}$ vs $1 \text{ mol L}^{-1} \text{ HCOOH}$ (methanoic/formic acid)
- Regardless of concentration, HCl is ALWAYS stronger than HCOOH

HIA Shen-Kit at 1/07/2020 1:26 PM

- **Electrolyte:** Solution that conducts electricity
- In an electrolyte, **ions** conduct electricity, not electrons

Lucarelli Ch 17 - Acids and Bases

Monday, 29 June 2020 1:33 PM

17.1 - Chemical and Physical Properties of Acids and Bases

- Acidic solutions turn blue litmus red, sour taste
- Basic solutions turn red litmus blue, bitter taste, slippery/soapy feel
- Salts are formed when one or more H^+ ions are replaced by metal cations
 - E.g. HCl salts: NaCl, KCl, $CaCl_2$, $MgCl_2$

Equations of Acidic Solutions

- Acid + Reactive Metal $\rightarrow$ Salt + $H_{2(g)}$
- Acid + Metal Hydroxide $\rightarrow$ Salt + $H_2O_{(l)}$
- Acid + Metal Oxide $\rightarrow$ Salt + $H_2O_{(l)}$
- Acid + Carbonate $\rightarrow$ Salt + $H_2O_{(l)}$ + $CO_{2(g)}$
- Acid + Hydrogen carbonate $\rightarrow$ Salt + $H_2O_{(l)}$ + $CO_{2(g)}$
- These reactions are a two-stage process:
 - Initially forms carbonic acid (H_2CO_3) which decomposes into $H_2O_{(l)}$ + $CO_{2(g)}$
- Acid + Metal Sulfite $\rightarrow$ Salt + $H_2O_{(l)}$ + $SO_{2(g)}$

Equations of Basic Solutions

- Base + Ammonium Salt $\rightarrow$ Salt + $H_2O_{(l)}$ + $NH_{3(g)}$
- Base + Non-Metal Oxide $\rightarrow$ Salt + $H_2O_{(l)}$

17.2 - Ionisation of Water and pH

- A tiny percentage of water molecules exist as H_3O^+ and OH^- ions (means same as H^+ and OH^- , but shows that the H^+ ion is attached to another H_2O molecule)
- This happens in a reversible reaction (rapidly convert between ions and molecules)
 - $H_2O_{(l)} \rightleftharpoons H^+_{(aq)} + OH^-_{(aq)}$
 - $2H_2O_{(l)} \rightleftharpoons H_3O^+_{(aq)} + OH^-_{(aq)}$
- Produces equal number of H^+ and OH^- ions, therefore pure water is neutral
- Concentration of H^+ and OH^- is $1 \times 10^{-7} \text{ mol L}^{-1}$
 - Corresponds to a pH of 7 (neutral)
- Dissolving an acid or base in water will increase the H^+ or OH^- concentration accordingly, changing the $H^+_{(aq)}$ concentration and therefore the pH

17.3 - The Arrhenius Theory of Acids and Bases

- Acid: A species which will ionise (acids have a covalent molecular structure) in solution producing $H^+_{(aq)}$ ions
 - $HNO_{3(aq)} \rightarrow H^+_{(aq)} + NO_3^-_{(aq)}$
- Base: A species which will dissociate (all ionic compounds (according to Arrhenius)) in solution producing $OH^-_{(aq)}$ ions
 - $NaOH_{(aq)} \rightarrow OH^-_{(aq)} + Na^+_{(aq)}$
- It is unable to explain the acid-base behaviour of many compounds, e.g. why Na_2CO_3 and K_3PO_4 are basic in nature, although they don't release OH^- ions

17.4 - Strong and Weak Acids and Bases

- Strong vs Weak: Refers to degree of ionisation
- Strong acids completely (or close to completely) ionise - almost every molecule will ionise
 - Shown using single arrow ($\rightarrow$)
- Weak acids partially ionise - only a small percentage of molecules will produce H^+ ions

- Shown using double arrow ($\rightleftharpoons$)
- Therefore for the same concentration, strong acids will produce a higher H^+ concentration than weak acids

17.5 - Acid-Base properties of Metal and Non-Metal Oxides

- Many non-metal oxides are acidic
 - Non-metal oxide combines with water to form an acid, acid dissolves releasing H^+ ions
 - E.g. $\text{CO}_{2(\text{g})} + \text{H}_2\text{O}_{(\text{l})} \rightleftharpoons \text{H}_2\text{CO}_{3(\text{aq})}$
 - $\text{H}_2\text{CO}_{3(\text{aq})} \rightleftharpoons \text{H}^+_{(\text{aq})} + \text{HCO}_3^-_{(\text{aq})}$
 - $\text{HCO}_3^-_{(\text{aq})} \rightleftharpoons \text{H}^+_{(\text{aq})} + \text{CO}_3^{2-}_{(\text{aq})}$
- Many metal oxides are basic
 - Metal oxide dissociates, releasing O^{2-} ions; O^{2-} combines with H_2O molecules to form OH^- ions
 - E.g. $\text{Na}_2\text{O}_{(\text{s})} \rightarrow 2\text{Na}^+_{(\text{aq})} + \text{O}^{2-}_{(\text{aq})}$
 - $\text{O}^{2-}_{(\text{aq})} + \text{H}_2\text{O}_{(\text{l})} \rightarrow 2\text{OH}^-_{(\text{aq})}$
 - Even some insoluble metal oxides will absorb water to produce a metal hydroxide
 - E.g. $\text{MgO}_{(\text{s})} + \text{H}_2\text{O}_{(\text{l})} \rightarrow \text{Mg}(\text{OH})_{2(\text{aq})}$

17.1 - Properties of Acids and Bases

- Solutions of acids/bases can conduct an electric current
- A soluble base is called an alkali
- Indicators are chemicals that change colour depending on the acidity of a solution
- More precise pH measurements can be made using a pH scale

17.2 - Arrhenius Model

Acids

- Acid: A species which ionises in aqueous solution to produce H^+ ions
- In general, only the hydrogen atoms bonded to a highly electronegative atom can dissociate from the molecule and ionise as the polar bond can be broken more easily
- Common properties of acids come from the presence of H^+ ions

Bases

- Base: A species which dissociates in aqueous solution to produce OH^- ions
- Common properties of bases come from the presence of OH^- ions

Limitations

- Can't explain why substances that don't contain hydrogen can form acidic solutions
 - E.g. CO_2 , SO_2
- Can't explain why substances that don't contain hydroxide ions can form basic solutions
 - E.g. NH_3 , $NaHCO_3$
- Restricted to substances that are soluble in water - can't explain acid/base behaviour in non-aqueous solutions

Strength vs Concentration

- Strong acids are completely ionised ($\rightarrow$)
- Strong bases are completely dissociated ($\rightarrow$)
- Weak acids/bases partially ionise ($\rightleftharpoons$)
- Concentration refers to the amount of acid/base dissolved in a given volume of solution
Usually in mol L^{-1}

pH

- $pH = -\log_{10}[H^+]$; $[H^+] = 10^{-pH}$
- H^+ = concentration of H^+ in mol L^{-1}
- Acidic: $pH < 7$
- Neutral: $pH = 7$
- Basic: $pH > 7$

17.3 - Reactions of Acids and Bases

- Acid + Metal Hydroxide $\rightarrow$ Salt + Water
- Acid + Metal Carbonate/Hydrogen-carbonate $\rightarrow$ Salt + Water + Carbon Dioxide
- Acid + **Reactive** Metal $\rightarrow$ Salt + Hydrogen Gas
- Base + Ammonium Salt $\rightarrow$ Salt + Water + Ammonia Gas

17.4 - Calculations Involving Acids and Bases

- Amount of solute (moles) doesn't change during dilution
- Change in concentration can be calculated using $c_1V_1 = c_2V_2$
- Concentration of an acid can be determined with **all** of the following:
 - Volume of concentrated solution
 - Molarity of concentrated solution
 - Total volume of water added
- When diluting an acid, its pH can never go above 7
- When diluting a base, its pH can never go below 7

Stoichiometry

Monday, 3 August 2020 6:45 PM

Calculations

- $n = \frac{m}{M}$
 - $m = nM$
- $n = \frac{V}{22.71}$
 - $V = 22.71n$
 - Only at STP (0°C and 100kPa)

Limiting Reagent

- $aX + bY \rightarrow \text{Products}$, where a and b are coefficients, and X and Y are elements
1. Calculate moles of each substance (if not given)
 2. Calculate the relative moles of one of them $\left(n(X) * \frac{b}{a} = rn\right)$
 3. If $rn(X) > n(Y)$, Y is the LR. If $n(Y) > rn(X)$, X is the LR
- E.g. $2\text{Al}_{(s)} + 6\text{HCl}_{(aq)} \rightarrow 2\text{AlCl}_{3(aq)} + 3\text{H}_{2(g)}$ (9.87mol Al added to 24.5mol HCl)
- $$\text{Relative moles of HCl} = n(\text{HCl}) * \frac{2}{6} = 8.167\text{mol}$$
- $8.167 < n(\text{Al})$, therefore HCl is the Limiting Reagent

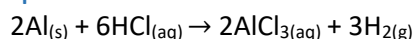
10.6 - Measuring Gases

- At STP, one mole of a gas occupies a volume of 22.71L
 - STP is 100kPa and 0°C
 - $n = \frac{V}{22.71}$; $V = 22.71n$
- Increasing pressure or decreasing temperature will make the gas occupy less space

11.7 - Determining the Limiting Reagent

- Find the moles of each substance
- Divide the mole of one over moles over the other (Actual Ratio)
- If the Actual Ratio > Stoichiometric Ratio (given by coefficients), the denominator is the LR, if $AR < SR$, numerator is the LR

Example



9.87mol Al added to 24.5mol HCl

1. Given

$$2. \quad AR = \frac{n(\text{HCl})}{n(\text{Al})} = \frac{24.5}{9.87} = 2.48$$

$$3. \quad SR = \frac{n(\text{HCl})}{n(\text{Al})} = \frac{6}{2} = 3 \text{ (in equation)}$$

$AR < SR$, therefore HCl is the limiting reagent

18.1 - Stoichiometry Involving Solutions

- Use $n = cV$ to find moles then use stoichiometry as usual

14.2 - Molar Volume of Gases

- Molar volume = 22.71L

Standard Conditions

- Standard Temperature and Pressure (STP): 0°C and 100kPa pressure
- Ideal gas: Theoretical gas composed of particles that do not interact except during collisions
 - At STP, most gases behave similar to an ideal gas, although they vary slightly
 - Stronger intermolecular forces = greater variation from ideal gas volume

14.3 - Calculations Involving Reactions with Gases

- $m(\text{reactants}) = m(\text{products})$
- $$\frac{n(\text{unknown})}{n(\text{known})} = \frac{\text{coefficient of unknown}}{\text{coefficient of known}}$$
- Use $n = \frac{V}{22.71}$ or $n = \frac{m}{M}$ to calculate moles, then use stoichiometry to find the unknown